

Engineering practice and automation

Every technique must discharge a named verification claim.

Make
responsibilities
observable



Read machine-
learning evidence



Keep accountable
human gates

Learning outcomes

Demonstrate these decisions on a real plan row.

- Assign ownership across verification handoffs.
- Interpret an ML result by task and denominator.
- Admit generated checkers through accountable review.
- Choose an improvement from the local verification bottleneck.

Route through the lesson

Four blocks, each ending with a decision.

- Make responsibilities observable
- Read machine-learning evidence
- Keep accountable human gates
- Choose the next improvement

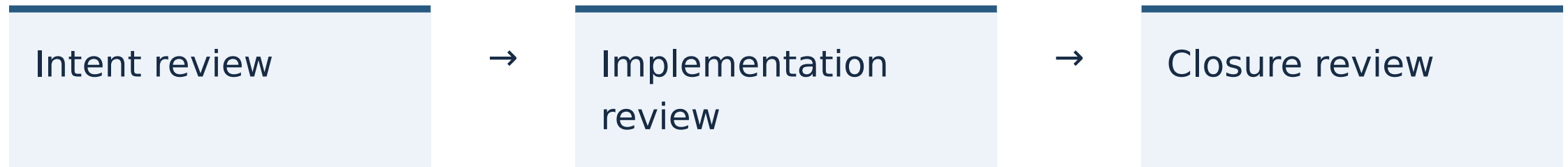
An artifact owner is not the whole outcome

Integration seams need explicit responsibility.

- Name who resolves cross-team discrepancies.
- Keep the expected evidence visible at handoff.

Reviews answer different questions

Plan, code and closure reviews need different inputs.



Reproduction is an onboarding milestone

A newcomer should establish evidence before explaining architecture.

- Start from a known failure and its reproducer.
- Produce a useful bug report and trace the ownership path.

Discussion: every deliverable has an owner

Name the evidence that would settle the decision.

- The model and checker were both delivered.
- Their disagreement blocks integration.
- Which responsibility was left unnamed?

Evaluate the proposed operational use

A measured task outcome is more useful than generic accuracy.

- Use held-back data and a relevant comparison.
- Re-measure when the design or failure population changes.

A benchmark needs its denominator

Artifact class and acceptance criterion determine the meaning.

- Record filtering and excluded cases.
- Compare like tasks before comparing headline results.

A generated checker can become a bad oracle

A proof against current RTL does not establish intended behavior.

- Inspect timing direction and activation.
- Challenge both design and generated interpretation.

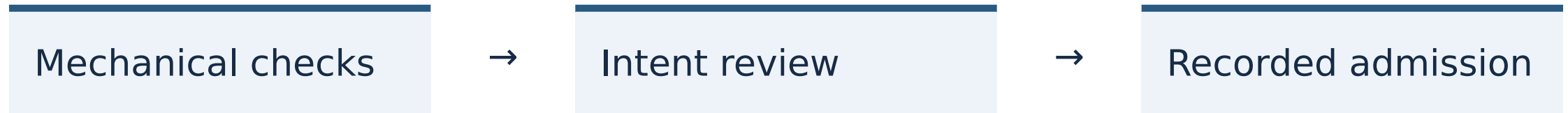
Discussion: most generated properties prove

Name the evidence that would settle the decision.

- Generated properties pass against the current RTL.
- No one has checked them against intent.
- What has the reported result established?

Filter mechanically before intent review

Human attention should reach candidates with a clear provenance.



Correlated errors defeat casual sampling

A shared generation mistake can affect an entire batch.

- Inspect systematic translation risks.
- Do not infer batch correctness from a few plausible examples.

Preserve the accepted artifact, not just the model name

Provenance must support reconstruction of the decision.

- Retain relevant inputs, generated text and accepted revision.
- Record reviewer, rationale and applicable restrictions.

Discussion: approval of a sampled batch

Name the evidence that would settle the decision.

- A reviewer checks a few generated properties.
- All share the same translation template.
- What uncertainty remains about the batch?

The bottleneck depends on the project level

A block and a software workload can need different improvements.

- Separate knowing what to check from execution capacity.
- Measure the local constraint before investing.

Earlier and later evidence are complementary

Moving a check does not erase its original scope obligations.

- Use early models with explicit fidelity limits.
- Retain deep block checking alongside system execution.

Automation moves work without removing judgment

Partial oracles and model boundaries remain engineering problems.

- Improve the weakest evidence-producing step.
- Keep interpretation, review and ownership explicit.

Discussion: cut regression after a new proof

Name the evidence that would settle the decision.

- A new formal result motivates removing simulation work.
- Its model covers only part of the old claim.
- What evidence is required before the cut?

Exercise: compare two automation proposals

Distinguish task evidence from design consistency.

- One classifier is measured against held-back human triage.
- Generated properties instead prove against current RTL.
- Explain which result measures its proposed use and what remains unknown.

What to change on Monday

Make one existing verification decision reviewable.

- Repair one unowned integration handoff.
- Audit provenance for an accepted generated checker.
- Measure the bottleneck before selecting an automation project.

Scope and next reading

Use the handout for the detailed artifacts and sources.

- Read the Engineering practice and automation handout and book Chapters 24–26.
- Model training recipes and predictions about future product performance are outside this lesson.